

School Name

Engineering Studies - Preliminary Course

Engineering Studies Task 2



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1 Abstract

This report details a use of applied mathematics in the calculation of various forces within a hypothetical vehicle. We take randomised values of a multitude of characteristics and measurements to display the intuitive application and derivation of formulas that may be extended to more complex models that better reflect real-world scenarios in the industry. We also undertake this report to showcase some potential considerations when making these calculations, particularly with respect to choices made in the inclusion and exclusion of certain parameters. A delegation of responsibilities is included at the end for convenient attribution.

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2 Introduction

Braking systems are no doubt a crucial element of the modern automotive, and their complex design is only made possible by the countless hours engineers and mathematicians dedicate to ensure the safety of everyone. Specifically, we combine the experimental calculation of the coefficient of friction, μ , of a brake pad against a steel bar with randomised vehicular characteristics to provide a comprehensive series of calculations to consider the stopping distance and braking force of the hypothetical vehicle at a predetermined speed. We additionally calculate a parking brake system to illustrate these calculations within a more static setting. Applied mathematics is a far more diverse field than as represented here, but still an invaluable tool for engineers nonetheless.

3 Brake Pad Coefficient of Friction

3.1 Calculating the coefficient

Consider the definition for the coefficient of friction:

$$\mu := \frac{F_{\max}}{N}$$

where $F_{\max}$ is the maximum frictional force, and N the normal force. Hence, consider the following diagram. Hence, consider that as the system is static, $f = mg \sin \theta$, and

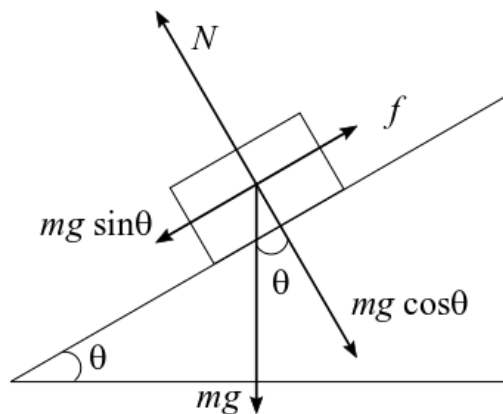


Figure 1: Aiden's diagram of an incline plane

that $\exists \theta : mg \sin \theta = F_{\max}$. Hence, the coefficient of friction is simply

$$\mu = \frac{mg \sin \theta}{mg \cos \theta} = \tan \theta$$

where θ is the angle of repose such that the object will tend to move. Hence, we record the angle of repose across a number of trials:

Trial number	Measured θ
1	28°
2	31°
3	26°
4	28.5°
5	28°
6	25°
Mean	27.75°

And the result is a trivial calculation:

$$\mu = \tan(27.75^\circ) \approx 0.58. \quad (2\text{DP})$$

3.2 Determining a radius for the brake pad

Consider that we want the average radius for the brake pad. As it is nontrivial to directly take radius from the arc, consider the following diagram:

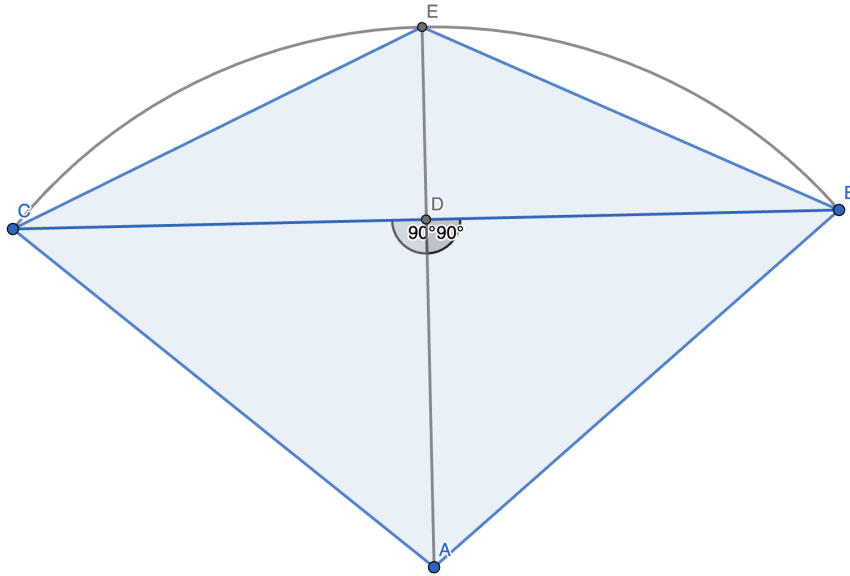


Figure 2: Rex's geogebra diagram for calculating the radius

As we are given the arc centred at A from B to C , draw the line BC and take D as its midpoint. Draw the line $DE \perp BC$. Consider, the radius is simply AE . Define the following:

$$r := AE$$

$$h := DE$$

$$c := BC.$$

Hence, by the Pythagorean theorem, we have:

$$r^2 = (r - h)^2 + \frac{c^2}{2}$$

$$r^2 = r^2 - 2rh + h^2 + \frac{c^2}{4}$$

$$r = \frac{h}{2} + \frac{c^2}{8h}.$$

And by a substitution of values, we have $r_{\text{in}} = 123$, $r_{\text{out}} \approx 170.8409$, and $r_{\mu} \approx 146.9205$, where r_{μ} denotes the arithmetic mean of the inner and outer radii.

4 Vehicle Foot Brake System

4.1 Vehicle characteristics

Define the following from the given table:

$$\text{Constants} := \begin{cases} g &= 5 \\ h &= 10 \\ i &= 22 \\ j &= 6 \end{cases}$$

Hence, we have the following:

Characteristic	Value
Mass	2053kg
Wheelbase	2.55m
Height of CM	0.553m
Dia. of wheel (incl. tyre)	625mm
Dia. of front calliper piston	50mm
Dia. of master piston	37mm
Dia. of booster diaphragm	166mm

4.2 Stopping Distance and Braking Force

4.2.1 Speed

It is given that $s = 0.1\text{SP} + 0.006\text{SP}^2$, where SP is the speed. Hence, $s \approx 58\text{m}$ for $\text{SP} = 91\text{km/h}$.

4.2.2 Kinetic Energy

It is well known that $\text{KE} = \frac{1}{2}mv^2$. Hence, take $v = \frac{455}{18}\text{m/s}$ and we have $\text{KE} \approx 656\text{kJ}$.

4.2.3 Braking Force

Consider $W = F \times d$. A simple rearrangement yields $F_s \approx 11.2\text{kN}$ by substituting values previously found.

4.3 Normal Reaction and Force on Tyres

4.3.1 Normal Reaction

Consider $N = \frac{m}{4g}$ for the normal force on each tyre (given there are 4). Hence, $N = 5132.5\text{N}$ per tyre. No moments need to be considered; none are induced.

4.3.2 Required and maximum friction force

Consider $F_F = \frac{F_s}{4} \approx 2710\text{N}$ for the required braking force on each tyre to stop within the required distance. The maximum friction force is trivially $5132.5 \times 0.9 \approx 4620\text{N}$.

4.3.3 Moment of the frictional force

It is known that the radius of the tyre and wheel combined are $r = 0.3125\text{m}$. Hence, the two moments are given as follows:

$$\begin{aligned}M_{\max} &= r \times F_{\max} \approx 1443. \\M_{\text{req}} &= r \times F_F \approx 872.\end{aligned}$$

4.4 Friction Forces and Piston Forces – Disk brakes

4.4.1 Required friction

Take the required moment from above and the distance of the point of application from the center to be r_μ . Hence, the frictional force is given by

$$F_{\text{disk}} = \frac{M_{\text{req}}}{r_\mu} \approx 5950\text{N}$$

for the force at each set of brakes (front and rear, left and right).

4.4.2 Necessary normal

Let us assume the previous force to find the necessary normal (equivalent to braking force) on each brake disk. We have already shown $\mu = \tan \theta$, and hence, $F = \frac{F_{\text{disk}}}{\mu} \approx 11.3\text{kN}$ for each brake disk.

4.5 Piston Forces and Hydraulic System

4.5.1 Front and rear cylinder

Assume a single cylinder applies the force. It is well known that $P = \frac{F}{A}$. Substitution evaluates to $P = \frac{F}{\pi \times 25^2 \times 10^{-6}} \approx 5.76\text{MPa}$ for the piston force. The rear piston must be of equal size to the front piston if the assumption that the frictional force necessary is equal.

4.5.2 Master cylinder

Using the same expression, we have $F \approx 6200\text{N}$ by rearranging with diameter 37mm.

4.6 Vacuum Booster System

Consider $\Delta F = \Delta P \times A$. Hence, $F = 50000 \times \frac{166^2}{2} \times \pi \times 10^{-6} \approx 1080\text{N}$.

4.7 Pedal Lever System

Consider the necessary force as $F - \Delta F \approx 5110\text{N}$. Hence, as the ratio is given, we have a mechanical advantage of 3.5 : 1 such that the necessary force is $F \approx 1460\text{N}$.

5 Vehicle Park Brake System

5.1 Brake Torque

By trigonometry we have the necessary torque as:

$$T = mg \sin 20^\circ \times \frac{0.17}{2} \times 4^{-1} \approx 37.3\text{Nm}$$

for each wheel. Friction force on each drum is also a trivial calculation:

$$F_F = \frac{mg \sin 20^\circ}{4} \approx 1760\text{N}$$

5.2 Drum Brake System

Take μ to 2 decimal points. Trivially, $F = \frac{F_F}{\mu} \approx 3030\text{N}$ for each drum. Halve this to 1515N for each shoe. It is known that $F = 7.6 \times T$ such that $T = 199\text{N}$.

5.3 Cable System

Consider:

$$T_{\text{handle}} = \frac{8T}{0.95} \approx 1682\text{N}$$

as the tension must be applied twice due to the dual shoes on 4 drums. Then:

$$L\varepsilon = \frac{T \times L^2}{A \times E} \approx 0.005\text{mm}$$

returns the expected elongation by multiplying strain with total length, where T denotes handle tension, L denotes total length, A denotes cross sectional area, and E denotes youngs' modulus.

5.4 Hand Brake Lever System

Define the following: $l_a := 226$, $l_b := 57$. From this, take the tension of the total cable as $T = 1682$. Hence:

$$F_{\text{hb}} \times l_a = F_b \times l_b \implies F_{\text{hb}} \approx 424.4\text{N}$$

and we have the required handbrake force.

6 Group Work Delegations

Task	Assigned + Evaluation
Overall typesetting	Anon: The formatting looked extremely nice, and by the end of the assessment task, other than a few minor tweaks, Anon did a sound job in organising and structuring the work.
Section 1	Anon: Anon collected the data while Anon handled the calculations and writeup in this regard.
Section 2.1	Anon: I designated and assigned the letters and numbers to both Student 1 and Student 2. Additionally, I completed the Table of Vehicle Physical Characteristics, hence completing the basic calculations required for each parameter. Finally, I formatted and designed this section. <i>Overall a great job here, a clear outline saves both of us a lot of time.</i>
Section 2.2-7	Anon: Quite clean, though maybe a bit difficult for non-technical readers to understand. Clearer explanations of mathematical notation would have been appreciated.
Section 3	Anon: Numerically correct with some silly errors later corrected. Quite clear otherwise, and easy to typeset in review. Units probably should have been better indicated to prevent confusion.